



CANDIDATE
NAME

CENTRE
NUMBER

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CANDIDATE
NUMBER

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9709/42

June 2026

1 hour 15 minutes

You will need: List of formulae (MF19)

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.
- Where a numerical value for the acceleration due to gravity (g) is needed, use 10 m s^{-2} .

- The total mark for this paper is 50.
- The number of marks for each question or part question is shown in brackets [].

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[Turn over

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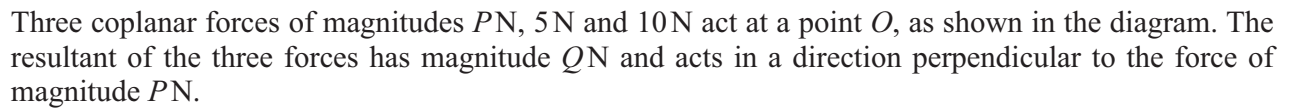
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- Find the tension in the rope.

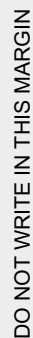
[4]

This image shows a full page of a document template. It consists of approximately 28 evenly spaced horizontal dotted lines across the entire width of the page, providing a guide for handwriting or typing. There are no margins, text, or other markings present.



[4]

This image shows a full page of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page, providing a template for handwriting practice or general writing. There are no margins, text, or other markings on the page.

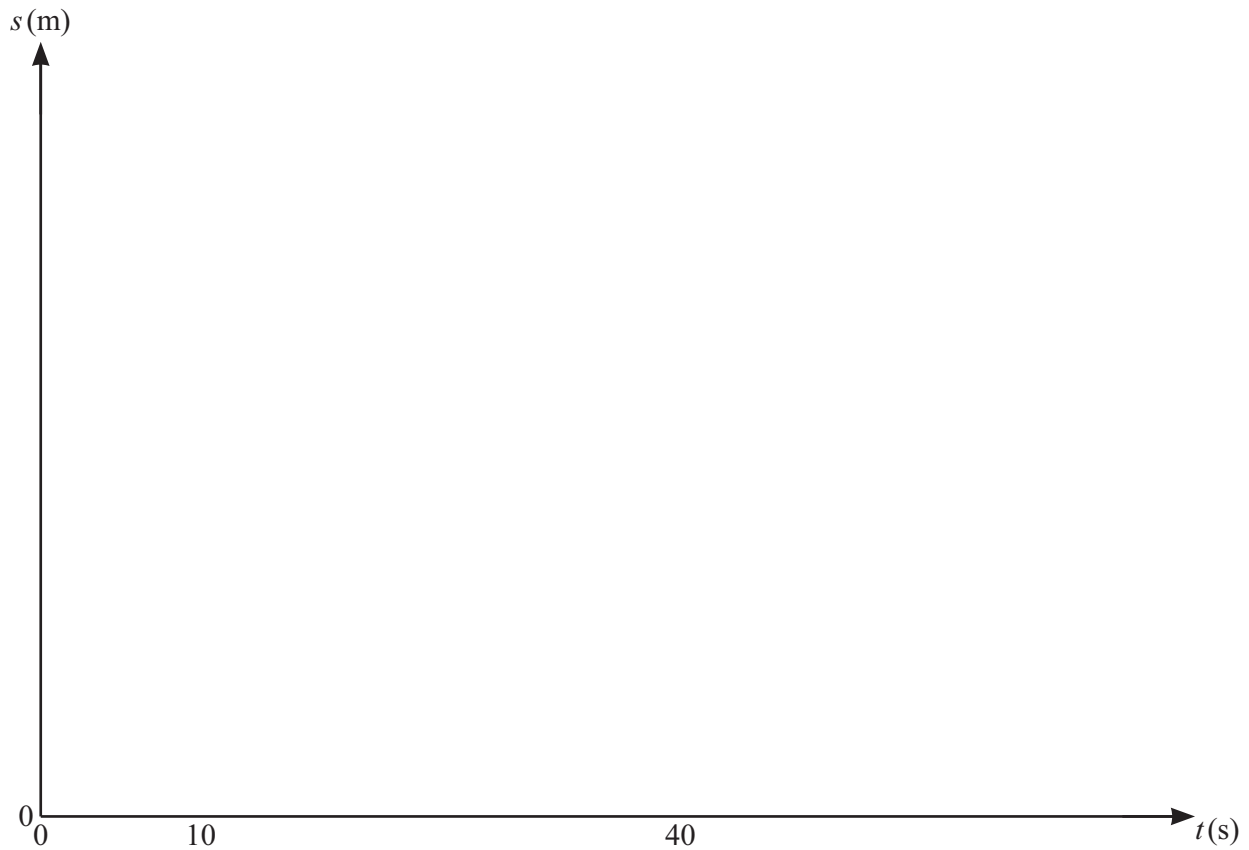


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- (b) On the given axes, sketch a displacement-time graph for the cyclist's journey from O to P , showing on your graph the distances travelled after 10 s and 40 s. [4]



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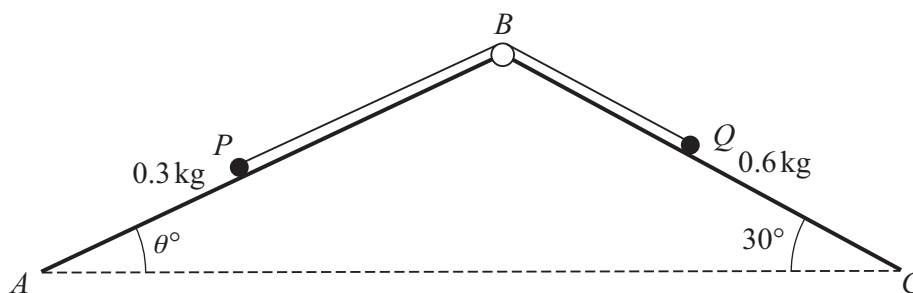
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It is given that the plane BC is smooth.

Find the tension in the string and the acceleration of Q .

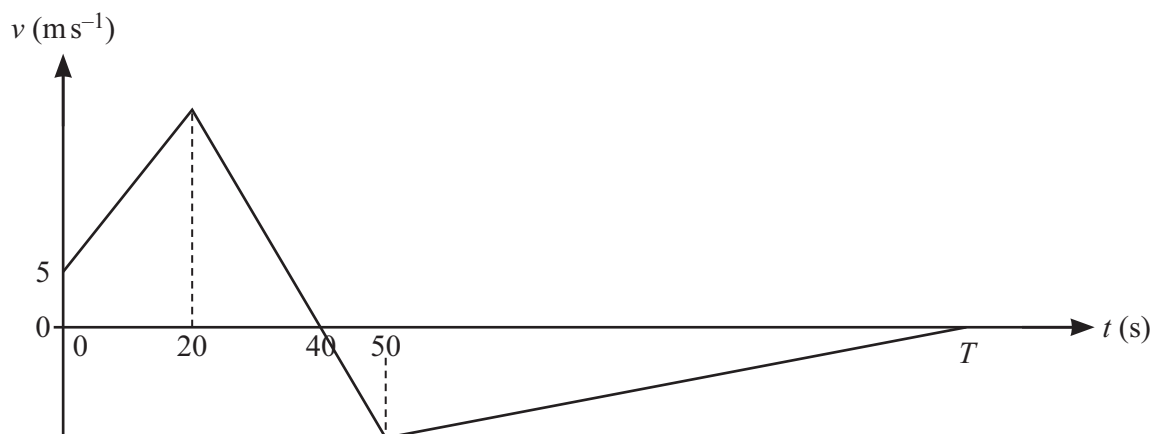
[5]

This image shows a full page of a handwriting practice worksheet. It consists of multiple sets of three horizontal dashed lines spaced evenly down the page, providing a guide for letter height and placement. The background is plain white, and there are no other markings or text present.

Find the value of X and the value of T . [4]

[illegible]

6



The diagram shows a velocity-time graph which models the motion of a particle. The graph consists of 3 straight line segments. The velocity of the particle at time t s after passing a fixed point O is $v \text{ ms}^{-1}$. The particle leaves O with a velocity of 5 ms^{-1} and accelerates at 0.75 ms^{-2} for 20 s. The particle then decelerates for the next 30 s. At $t = 40$, the velocity of the particle is zero. After $t = 40$, the particle starts to travel back to O , coming to rest at O at time T s.

(a) Find the value of T .

[5]

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(b) Find the acceleration of the particle from $t = 50$ to $t = T$.

[2]

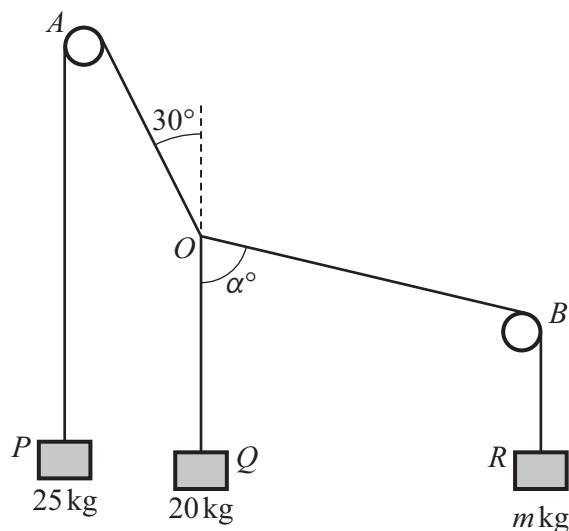
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Find the value of m and the value of α .

[6]

[illegible]

- (a) Find the velocity of X at $t = 4$. [2]

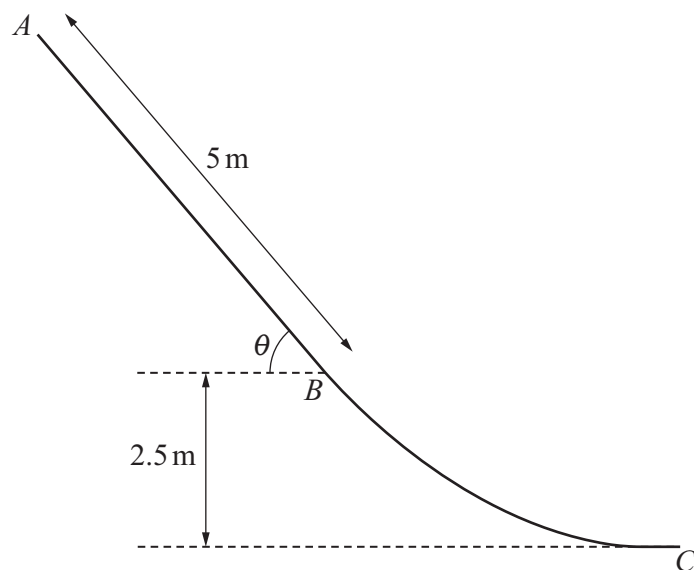
[illegible]

(b) (i) Find the value of T . [4]

This image shows a single sheet of white paper with horizontal blue ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

[4]

This image shows a full page of white paper with horizontal dotted lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.



The diagram shows the vertical cross-section ABC of a rough waterslide. The section AB is a straight line of length 5 m inclined at an angle of θ to the horizontal, where $\sin \theta = 0.8$. The point B is 2.5 m above the level of C . A man of mass 80 kg, modelled as a particle, slides down the waterslide, starting from rest at A . The coefficient of friction between the man and the straight section of the waterslide is 0.1.

Find the speed of the man at B .

[5]

[illegible]